

Q. 1. Build the analysis-ready comment corpus

Load your cleaned comments dataset into a pandas DataFrame. Check that the cleaned token column is usable for analysis, and convert each row into a single text string that can be used for machine learning. Remove rows with missing or empty text after conversion. Then print:

1. the number of usable comments
2. the name of the text field you used
3. one sample processed comment text

Answer. The cleaned comment tokens were checked first and then joined into one text column named `processed_text`. Empty rows were removed after conversion so that only usable comments remained for machine learning.

Result summary:

Number of usable comments	734
Text field used	<code>processed_text</code>

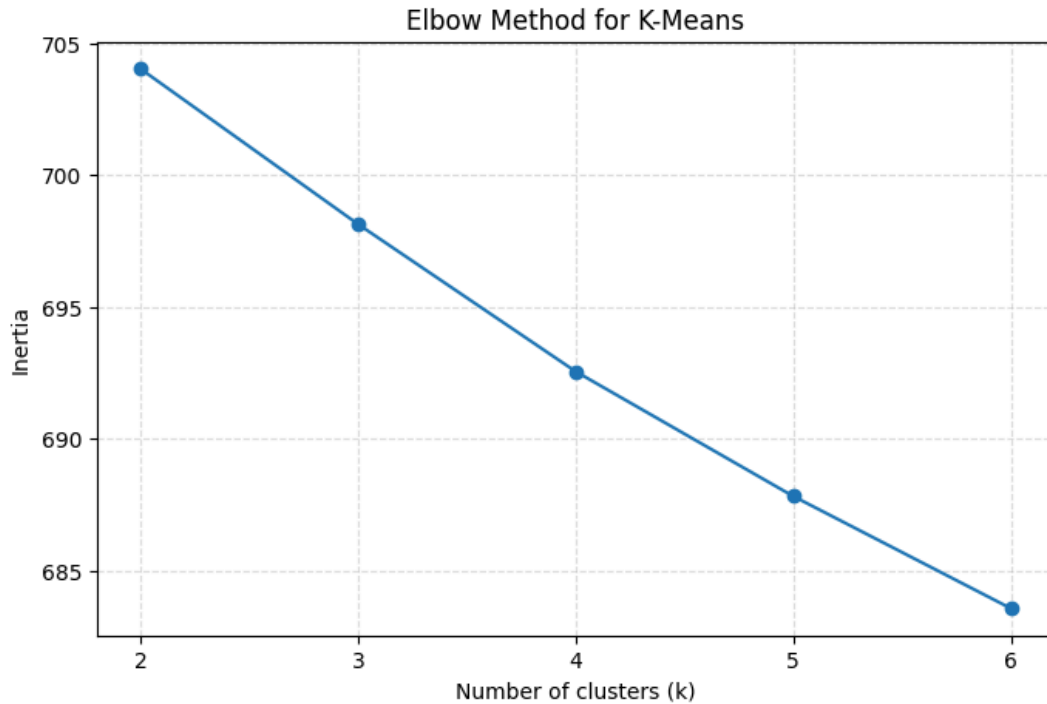
Sample processed comment: feeling lonely sky pick direction wave universe infinite means feeling lonely waving directly

Q. 2. Vectorize the comments and choose a suitable number of clusters

Convert the processed comments into TF-IDF features using a reasonable feature limit such as 1500. Then apply the Elbow Method for $k = 2$ to 6 and plot the inertia values. Based on the plot, choose one value of k and justify your choice in 1 to 2 sentences.

Answer. The processed comments were converted into TF-IDF features and the Elbow Method was tested for $k = 2$ to 6. After checking the inertia values and the elbow curve, $k = 4$ was selected as the most reasonable choice for the final clustering step.

Chosen k : $k = 4$



I selected $k = 4$ because the elbow plot shows a noticeable slowing in the drop of inertia after this point. The decrease continues for higher values, but the improvement becomes smaller, so four clusters gives a fair balance between structure and simplicity.

Q. 3. Group the comments and interpret the cluster themes

Run K-Means using your chosen value of k . For each cluster:

1. report the number of comments in the cluster
2. print the top 5 representative terms

Then write a short interpretation for any two clusters, explaining what kind of audience discussion each cluster seems to represent.

Answer. K-Means clustering was then applied using $k = 4$. For each cluster, the number of comments and the top representative terms were checked to understand the main discussion theme inside that group.

Cluster 0 seems to represent comments that are focused on the general science discussion in the video. The main words suggest that viewers in this cluster are

talking about space, the universe, time, and the idea of an edge. Cluster 3 looks more like an emotional or reaction-based group. Words such as brain, love, light, and infinity suggest that many comments in this cluster are about wonder, confusion, amazement, or personal reactions to the video topic.

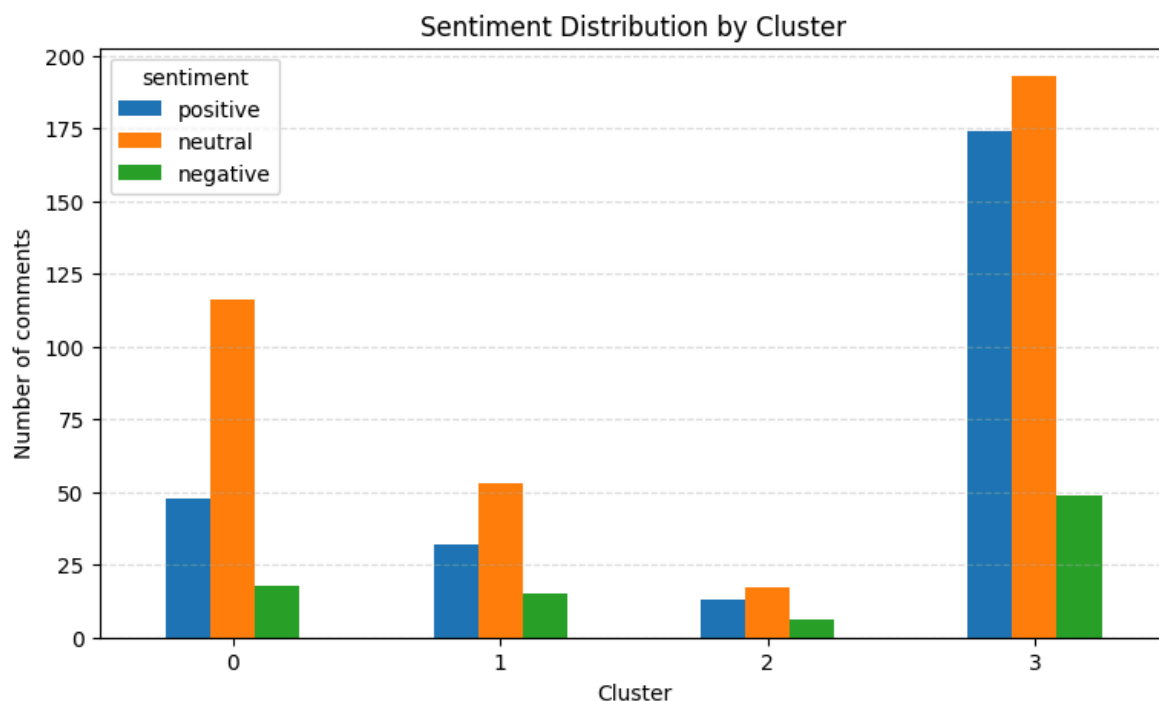
Q. 4. Compare sentiment across the discovered groups

Use a sentiment analysis tool such as TextBlob or VADER on the comments. For each cluster, calculate how many comments are positive, neutral, and negative. Present the result as either:

1. a table, or
2. a bar chart

Then identify which cluster appears most positive or most negative, and state one possible reason.

Answer. Sentiment analysis was used to count how many comments in each cluster were positive, neutral, and negative. This made it easier to compare the tone of discussion inside each cluster.



Cluster 3 appears to be the most positive overall because it has the highest number of positive comments. One possible reason is that this cluster contains many emotional reactions, appreciation comments, and excited responses from viewers.

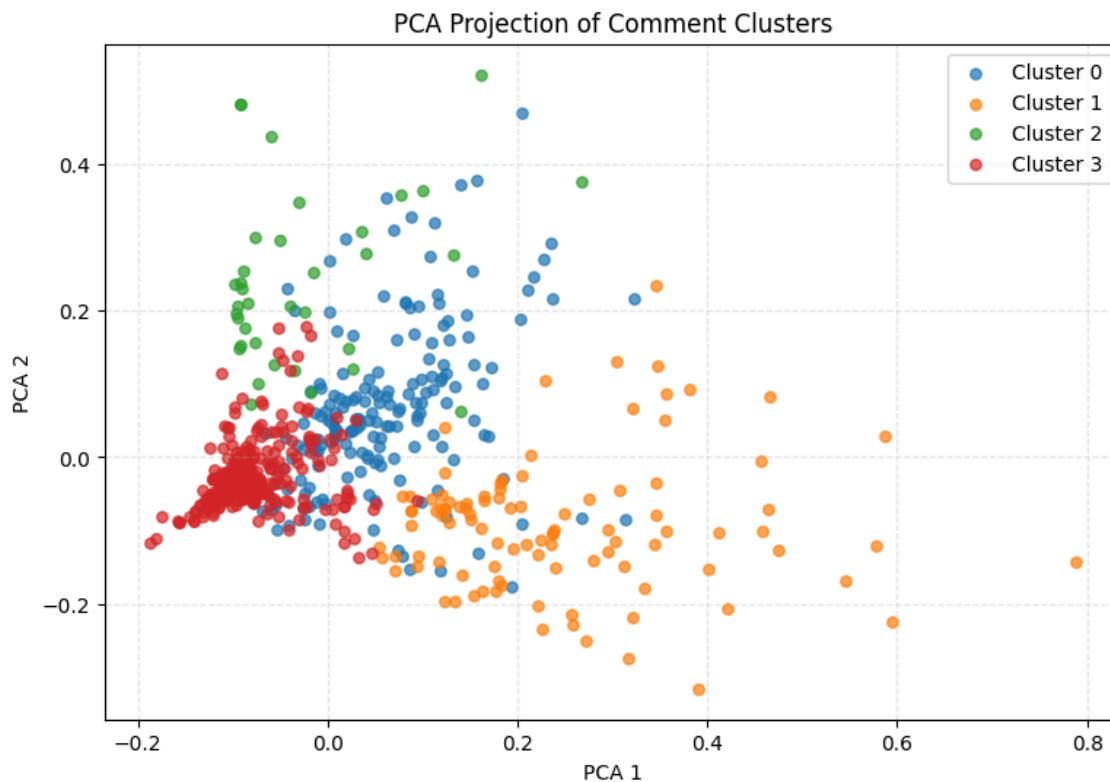
Cluster 1 looks slightly more mixed and relatively less positive compared to the large reaction cluster, because its main discussion is more about debating ideas such as whether the universe is infinite or finite.

Q. 5. Visualize the cluster structure and write a final observation

Reduce the TF-IDF feature space to 2 dimensions using PCA and create a scatter plot of the comments colored by cluster label. After plotting, write a short final observation of 3 to 4 sentences covering:

1. whether the clusters look clearly separated or partially mixed
2. whether the visual pattern supports your chosen value of k
3. one practical limitation of clustering short YouTube comments

Answer. PCA was used to reduce the TF-IDF feature space into two dimensions so that the clusters could be plotted on a scatter graph. This final plot helps show whether the clusters are clearly separated or still partly mixed.



The PCA plot suggests that the clusters are only partially separated, not fully distinct. This still supports the use of $k = 4$, because there is some visible grouping, but the overlap also shows that many comments share similar vocabulary. That is common in short YouTube comments, where people often use a few repeated words even when the meaning is different. A practical limitation of this kind of clustering is that very short comments do not always contain enough text to form a strong and clearly separate topic group.

In conclusion, the final results show that the comments can be grouped into a few broad themes, including science discussion, concept debate, and emotional viewer reactions. The sentiment analysis also shows that the comment section is mostly neutral to positive, with only a smaller portion of negative comments.